

1 Solution — GIAM 7.2

Problem 3

1.1 Problem

Provide an argument as to why an 8×8 chessboard with two squares pruned from diagonally opposite corners cannot be tiled with dominoes.

1.2 Setup

A standard chessboard is an 8×8 grid of unit squares, alternately colored *light* and *dark*. Remove two squares at diagonally opposite corners — say the corners **a1** (bottom-left) and **h8** (top-right), or equivalently **a8** (top-left) and **h1** (bottom-right). We ask whether the remaining 62-square region can be tiled by 1×2 dominoes, each placed horizontally or vertically, covering two adjacent squares.

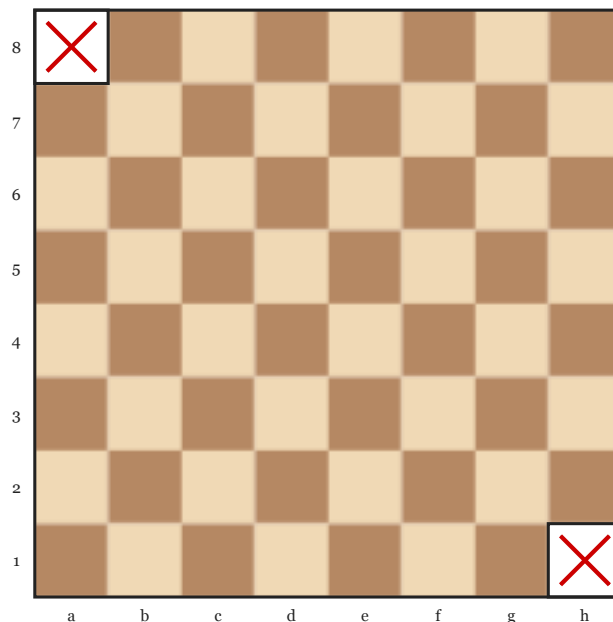


Figure 1.1: An 8×8 chessboard with two diagonally opposite light squares (top-left a8 and bottom-right h1) removed. The remaining 62 squares contain 32 dark squares and 30 light squares — an imbalance of 2 that obstructs any domino tiling.

1.3 Two Key Observations

Index each square by its position (i, j) , $i, j \in \{1, 2, \dots, 8\}$, and color it according to the parity of $i + j$:

$$\text{color}(i, j) = \begin{cases} \text{dark} & \text{if } i + j \text{ is even,} \\ \text{light} & \text{if } i + j \text{ is odd.} \end{cases}$$

This is the familiar checkerboard coloring; the full 8×8 board has **32** dark and **32** light squares.

1.3.1 Observation 1 — The Two Removed Corners Share a Color

The four corner squares of the chessboard sit at $(1, 1)$, $(1, 8)$, $(8, 1)$, $(8, 8)$, and the sums $i + j$ are

$$1 + 1 = 2, \quad 1 + 8 = 9, \quad 8 + 1 = 9, \quad 8 + 8 = 16.$$

So the *diagonally* opposite pairs $\{(1, 1), (8, 8)\}$ and $\{(1, 8), (8, 1)\}$ each have *matching* $i + j$ parity — both even or both odd. The pairs across the *anti*-diagonal $\{(1, 1), (8, 8)\}$ are both dark; the pairs across the other diagonal $\{(1, 8), (8, 1)\}$ are both light.

Either way, **the two corners we remove are the same color**. After removal, one color loses *two* squares while the other loses none. The new counts are

$$30 \text{ squares of the removed-corner color,} \quad 32 \text{ squares of the other color.}$$

1.3.2 Observation 2 — Every Domino Covers One Dark and One Light Square

A domino occupies a pair of *adjacent* squares — either (i, j) and $(i + 1, j)$ (vertical) or (i, j) and $(i, j + 1)$ (horizontal). In either case, the two cells' values of $i + j$ differ by exactly 1, so they have opposite parity, hence opposite color.

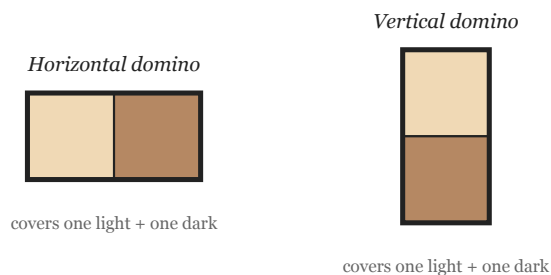


Figure 1.2: A horizontal or vertical domino on the chessboard always covers exactly one light square and one dark square, because two adjacent cells differ in $(i+j)$ by exactly 1.

So a tiling of any region by N dominoes covers exactly N dark squares and N light squares — *equal counts*.

1.4 The Argument

Suppose, for contradiction, that the mutilated 62-square board *could* be tiled by dominoes. Each domino covers 2 squares, so the tiling uses

$$N = \frac{62}{2} = 31$$

dominoes. By Observation 2, the tiling then covers 31 dark and 31 light squares.

But by Observation 1, the mutilated board contains 30 squares of one color and 32 of the other — neither of which equals 31.

Contradiction. Hence **no domino tiling of the mutilated board exists.** ■

1.5 Remark — A Parity Argument in Disguise

The proof is a parity argument at its core: it converts a *quantitative* fact (light–dark imbalance of size 2) into a *structural* impossibility (no tiling). What looks like a geometric question is really a counting question once the right invariant — the colors of the squares — is identified.

This is also the simplest case of a general principle: in a bipartite graph G with parts of unequal size, no *perfect matching* exists. Encode the mutilated board as a graph G whose vertices are the 62 remaining squares and whose edges connect horizontally or vertically adjacent squares. The dark and light squares form the two parts of a bipartition; a domino tiling is exactly a perfect matching of G . Since the two parts have sizes 30 and 32, no perfect matching exists.

1.6 Remark — The Classical Problem

This is the **mutilated chessboard problem**, popularized by Max Black in *Critical Thinking* (1946) and later by Martin Gardner, John McCarthy, and others. It is a touchstone example of how a clever coloring can render an apparently combinatorial-explosive search problem trivial: rather than enumerate possible tilings — there are astronomically many to check or rule out — the coloring argument settles the question in a single line of counting.

The same coloring trick generalizes broadly:

- **Removing two squares of *opposite* colors** can sometimes leave a tileable region — and in fact, by *Gomory’s theorem*, the result *can always* be tiled provided the two removed squares are of opposite colors.
- **Other shaped pieces** require richer invariants — e.g., tilings by L -trominoes or T -tetrominoes need finer colorings (sometimes with 3 or 4 colors), and the same parity-style impossibility proofs apply.

1.7 A Sharper Restatement

Putting the argument as a one-line lemma: *for any region $R \subseteq$ chessboard, if R admits a domino tiling, then R contains equal numbers of dark and light squares.* The mutilated board fails this necessary condition with strict inequality $30 \neq 32$, so it cannot be tiled.

(The converse — “equal counts imply tileable” — is *false* in general: there are regions with equal dark/light counts that still cannot be tiled. Our problem only needs the necessary direction.)